

Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
 - Bond, the present value of its coupons and face value
 - Stock, the present value of its expected dividends

Every Financial Asset Is Debt or Equity

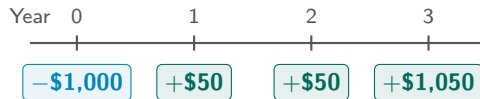
- **Financial asset**, value from a contractual right or ownership claim



Debt's Cash Flow Is Fixed, a Stock's Is Not

Debt

\$1,000, 3 years, 5% interest

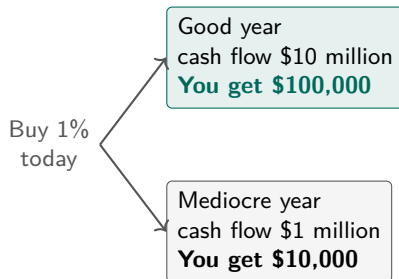


Every payment known in advance

Stock

1% of the company

Next year



Value of claim depends on how the company does

Common Types of Corporate Financing

Debt, lending to the company

Main forms, bonds and loans

Holder gets a fixed promised amount

“Senior”, paid back before equity in
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No voting rights or control

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Equity, stock ownership in the company

Holder gets dividends

“Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout

Voting rights

Module 5, Part 1

Bonds

Bonds, Definitions

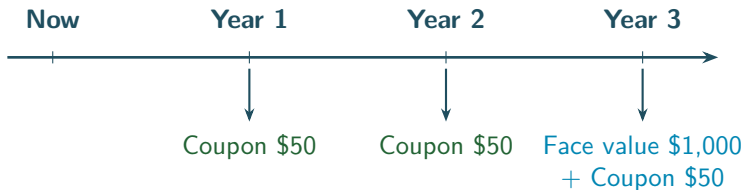
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- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$50 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.



Imaged by Heritage Auctions, HA.com

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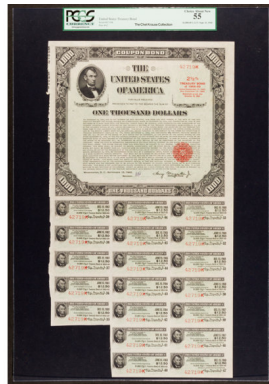


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- **Time to maturity**, the number of years until maturity
- **Coupon**, the periodic payments made to the bondholder before maturity
- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$



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Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate

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 - E.g., **Module 4**'s Puerto Rico bond, the coupon was 0 but the interest rate was 5.15%

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



Example, Puerto Rico Borrows

Recall from **Module 4**



$$N = 47, \quad I = ?$$

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- Solution,
 - 47 **N**
 - -94.40 **PV**
 - 0 **PMT**
 - 1000 **FV**
 - **I/Y**

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 - “Required rate of return”, the name for the discount rate in this application

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- E.g., for a bond with time to maturity T ,

$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

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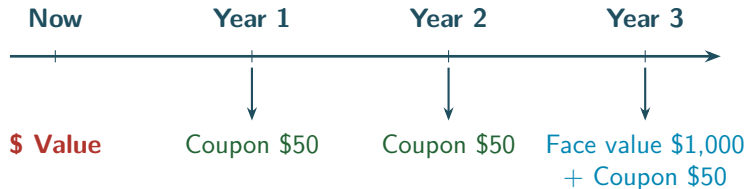
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- Investors choose the opportunity cost, “what could I get investing in something else with a similar risk”, as the required rate of return r

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$$\text{Value} = \frac{\$50}{(1 + 0.04)^1} + \frac{\$50}{(1 + 0.04)^2} + \frac{\$50 + \$1,000}{(1 + 0.04)^3} = \$1,027.75$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond

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 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond
- Solution,
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 - **CPT**, then **PV**
- The output is $-\$1,027.75$ (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
 - **N**, the number of periods (years) until maturity
 - **PV**, the value of the bond (the present value of the coupons and the face value)
 - **PMT**, the coupons (in dollars)
 - **FV**, the face value (the lump sum paid at the end)
 - **I/Y**, the discount rate (the required rate of return r)

Recap of Part 1

- Companies issue bonds (and other debt) to raise financing from investors

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- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors
- The value of a bond is the discounted payments, at the required rate of return

Module 5, Part 2

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- **Yield**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price

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- **Yield**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price
- Given the price of a bond (PV), its maturity T , and its coupons, find the yield by solving for y ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
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- **Here we are calculating the yield!**



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Example, Bond Yields

- Solving for the yield analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?

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- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?
- Solution,
 - 1000 **FV**
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- The yield is 4%

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r , output the value
(**CPT** **PV**)

Yield y

Required Rate of Return vs. Yield

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Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield
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Required Rate of Return vs. Yield

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- Input r , output the value
(**CPT** **PV**)
- **Subjective**

Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield
(**CPT** **I/Y**)
- **Objective**

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

Bond	Price	Maturity (years)	Yield
A	\$310	25	?
B	\$310	?	9%
C	?	11	8%

- What is the yield of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	−310	0	1,000
b. Bond B	?	9	−310	0	1,000
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Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield of bond A is 4.796%

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- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

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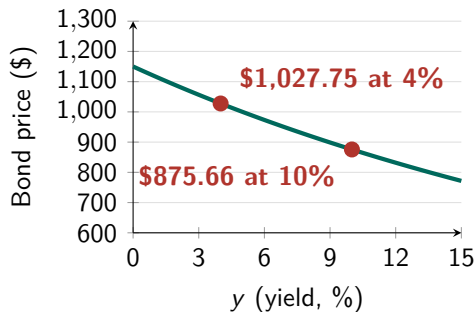
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c. Bond C	11	8	428.88	0	-1,000

- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

Bond Prices and Yields Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

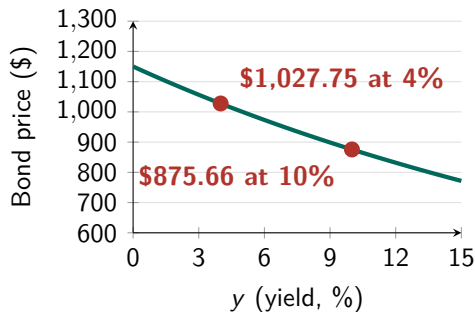


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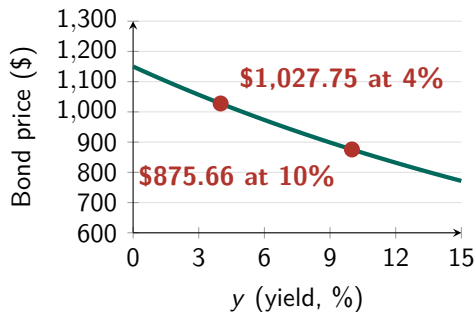


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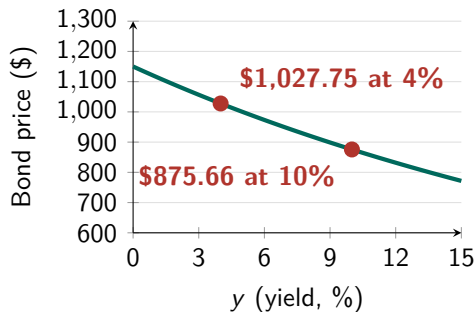


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 2. Risk of default (compensation for taking risks)
 - Higher risk of default, lower price, higher yield

Default Risk

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- **Default premium**, the additional yield on a bond that investors require for bearing default risk

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- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr
Investment-grade bonds		
Aaa	AAA	0.1%
Aa	AA	0.7%
A	A	2.2%
Baa	BBB	3.5%
High-yield bonds		
Ba	BB	15.7%
B	B	35.5%
Caa, Ca	CCC, CC	48.9%
C	C	n/a

Source, Brealey, Myers, and Marcus, Table 6.2

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- Yields tend to be higher (prices lower) for lower-rated bonds

Recap of Part 2

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- Yield, the discount rate for which the present value of the bond's payments equals its price
- Bond prices and yields move in opposite directions
- Prices fall and yields rise with market interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 3

Stocks

Goals

- **Part 1, bonds**
 - Differences between debt and equity
 - Valuation
- **Part 2, bond yields**
 - Yields
 - Interest rates
 - Default risk
- **Part 3, stocks**
 - Valuation, dividend discount model
 - Valuation, comparables

Valuing Stocks

- Two alternative ways to value stocks,
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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

Stock Valuation, Terminology

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- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**, $\frac{\text{Stock price}}{\text{Earnings per share}}$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?

Valuing Stocks, Dividend Discount Model

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- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?
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- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$ are *expected* dividends

Gordon Growth Model

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$$P_0 = \frac{\text{Div}_1}{r_e}$$

- If dividends are expected to grow at a constant rate g ,

$$P_0 = \frac{\text{Div}_1}{r_e - g}$$

- The latter formula is often called the “**Gordon growth model**”

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?

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- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?
- Solution, apply the perpetuity formulas,

$$1. \quad P_0 = \frac{\text{Div}_1}{r_e} = \frac{\$3.00}{0.12} = \$25$$

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?
- Solution, apply the perpetuity formulas,

$$1. \quad P_0 = \frac{\text{Div}_1}{r_e} = \frac{\$3.00}{0.12} = \$25$$

$$2. \quad P_0 = \frac{\text{Div}_1}{r_e - g} = \frac{\$3.00}{0.12 - 0.08} = \$75$$

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- Stock value today, the present value of these cash flows,

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- Rewrite the one-period equation as the expected return from the one-year investment,

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- That is, dividend yield plus expected capital appreciation (in %)

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- How to value non-dividend-paying stocks?
 - If the firm is only temporarily not paying dividends, the dividend discount model still applies
 - But future dividends are then probably even more uncertain!

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \cdots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

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$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

Practice Problem

- Consider the following three stocks,
 - a. Stock A is expected to pay a dividend of \$10 a share forever
 - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
 - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$P_C = \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} +$$

$$\frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6}$$

$$P_C = \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6$$

$$P_C = \$104.51$$

At a 10% capitalization rate, Stock C has the largest present value.

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At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity, $P_A = \$142.86$, $P_B = \$166.67$, and $P_C = \$156.50$, so Stock B is the most valuable

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- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

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- E.g., Company A creates twice the cash flow per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?
- Underlying idea, we can “scale” prices!

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- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Example, Valuation by Comparables

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- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

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- The question asks for the share price, not the market cap,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

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- Assumes the comparables are priced correctly
 - Cannot tell if an entire industry is overvalued or undervalued
 - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

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Thank you!

Please let me know if you have any questions.